Notes on Block Selection

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1 Introduction

Sparse attention reduces the quadratic cost of long context models to a manageable level. Our contributions are summarised in the following list.

- A block sparse attention kernel
- An analysis of the memory traffic
- A scheduler for mixed block sizes
- Experiments on six public corpora
- An open source reference implementation

2 Notation

We use the following notation throughout the paper.

Block a contiguous span of keys

Pattern the set of blocks that a query attends to

Budget the number of blocks per query

Router the module that selects blocks

Kernel the fused implementation on the accelerator

3 Related work

The original proposal is stated in the following terms, which we quote in full.

> attention is all you need
> and memory is all you have
> so spend it where the mass is
> and nowhere else at all
> that is the whole of the method

4 Results

The sparse kernel reaches the same accuracy as the dense baseline on every task that we tried.